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Sec - A

1. (b) Data and a pointer to the next node
2. (c) NULL
3. (b) self-referential structure
4. (b) head = = NULL
5. (b) newnode -> next = head; head = newnode;
6. (b) temp = head; head = head -> next; free (temp);
7. (c) traversing from tail to head
8. (b) n

9. (c) copying data of P->next into P, then deleting P->next

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10. (b) while (P != NULL)
11. (b) P->next == NULL
12. (c) Two pointers at different rates
13. (c) Three
14. (b) Exactly one node
15. (b) P = (struct node*) malloc (sizeof (struct node));
16. (b) Avoid special-case handling for insertion and deletion at the end.
17. (c) The first node

18. (b) It does not support random (indexed) access,
19. (b) Loss of access to all nodes, causing a memory leak

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20. (b) A slow pointer and a fast pointer
21. (c) The last node
22. (b) Nodes may be scattered anywhere

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23.

```
Struct node
{
  int data;
  Struct node *next;
};
```

24.

Given list:

10->20->30->40->NULL

The program adds each node with the next node:

- 10+20=30
- 20+30=50
- 30+40=70

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Output:- 30 50 70

25.

```
void insertEnd (struct node **head, int x)
{
  struct node *t;
  t = (struct node*) malloc (sizeof (struct node));
  t->data = x;
  t->next = NULL;
  if (*head == NULL)
  {
    *head = t;
    return;
  }
  struct node *p = *head;
  while (p->next != NULL)
  {
    p = p->next;
  }
  p->next = t;
```

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26.
```
Struct node* reverse (struct node * head)
{
  Struct node* prev = NULL;
  Struct node *current = head;
   Struct node * next ;
   while (current ! = NULL)
    {
     next  = current ->next;
     current ->next = prev;
     prev = current;
     current =next;
    }
    return prev;
    }}
```

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27
```
void deleteNode (struct node *p)
  {
     struct node * temp;
     temp = p ->next;
     p->next = temp->next;
      free (temp);
    }
```

Suppose we have: 40->20->30->NULL
If we want to delete 30, make 20->40 directly to 40.
So, 20->30->40->20....
The code can be: temp = p->next;

28. Initial list:

5->8->12->3->NULL

i) Insert 7 at the beginning:

7->5->8->12->3->NULL

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ii) Delete 12:-

7->5->8->3->NULL

iii) Insert 9 at the end:

7->5->8->3->9->NULL

Final list:-

7->5->8->3->9->NULL